AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

# Particulate Matter
## Monitoring & Health Effects

Issue 2026-10-11 | Entry window 11 October 2026 (full day) | 1 records screened in scope

| 1 | 1 | 2 | 0 | 0 |
|---|---|---|---|---|
| RECORDS IN SCOPE (11 EXCLUDED) | OF 10 SUBTOPIC BANDS POPULATED | EFFECT ESTIMATES WITH A CI | TIER-A RECORDS | RECORDS CARRIED ON METADATA ALONE |

## Signal of the day

**A one-record Sunday. The single admitted paper puts $PM_{2.5}$ against incident fatty liver disease at HR 1.19 (1.10–1.30) and claims a 700-protein signature mediates 15 % of it — with a confidence interval from 2.7 % to 37.7 %.**

52,190 participants free of metabolic dysfunction-associated steatotic liver disease at baseline, 624 incident cases over a median 11.8 years. Annual average $PM_{2.5}$ gave **HR 1.19** (1.10–1.30); an elastic-net signature of **700** $PM_{2.5}$-associated proteins gave the same **1.19** (1.08–1.31), enriched in NF-$\kappa$B signalling and cytokine–receptor interaction. Mendelian randomisation supported CD28, CSF3 and HAVCR2 as causally associated with MASLD, and structural equation modelling put the mediated fraction at **15 %** (2.7–37.7 %). **Where it is weak:** a 700-protein signature trained and tested in one cohort absorbs variance from everything $PM_{2.5}$ co-varies with, so 15 % is an upper bound on mediation rather than an estimate of it, and the interval spans more than an order of magnitude. The MR arm establishes protein-to-disease causality, not exposure-to-protein. Exposure itself is an annual mean read off an interpolated surface — the weakest link in the chain, and the one a measurement network addresses directly.

## What changed today

- **A single-record day, and that is the finding.** Eleven candidates were screened, one admitted. The rejected eleven are heatwave mortality, pesticide exposure, indoor radon, two records already carried in the 10 October issue and six journal-sweep records with no PM content — a Sunday entry date with one PubMed health hit and no instrumentation deposits at all.
- **Two estimates, one cohort, one exposure.** The exposure HR and the protein-signature HR are not independent findings: the signature is derived *from* $PM_{2.5}$ in the same participants, so its agreement with the exposure estimate is close to arithmetic rather than confirmatory.
- **Mediation reported as a point estimate, read as a range.** 15 % with a 2.7–37.7 % interval is compatible with a trivially small mechanism and with a dominant one. The abstract's framing — that the signature "mediated 15 %" — is the kind of precision this digest records but does not repeat as a finding.

### Window continuity

The 10 October issue closed with `last_entry_date` at 10 October. This issue collects **11 October in full**, continuous with it, and brings the watch current: the backlog that opened on 9 October is closed. The **W41 weekly** (4–10 October) is filed under 10 October. 12 October is not opened — under the 22:00 rule a run must not harvest the day it executes on before 22:00 local, and this run began at 17:17 local.

## Corpus analytics

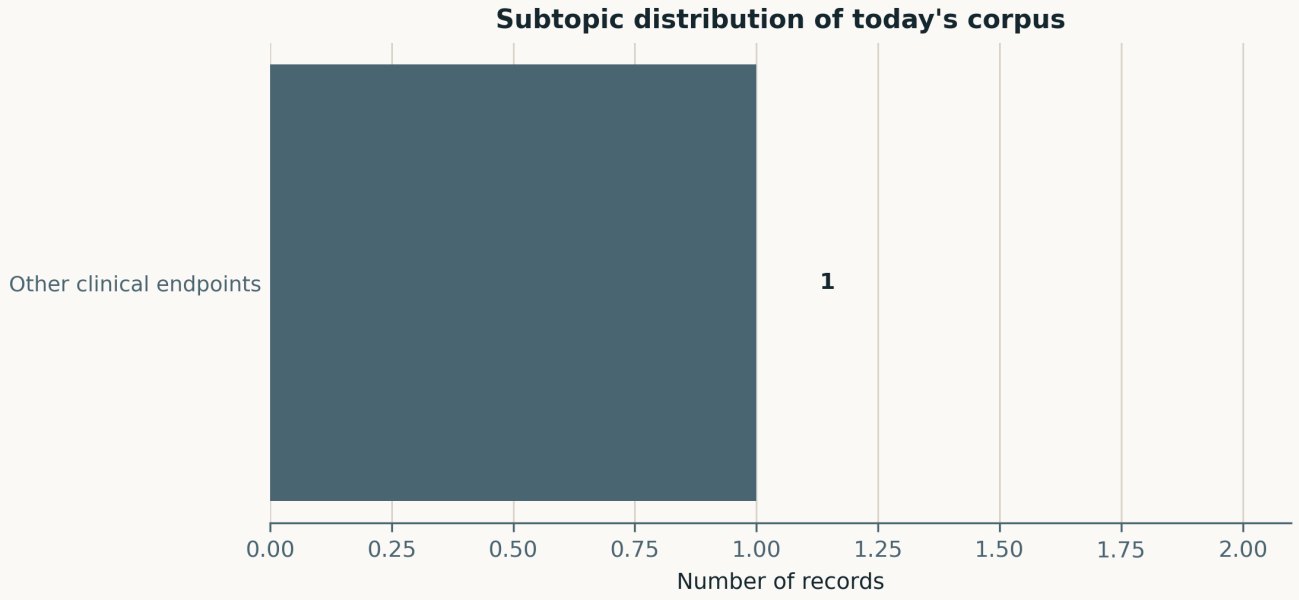



**Fig.** Subtopic assignment of the single in-scope record: *Other clinical endpoints*, one of ten bands populated. This is the smallest issue in the history of the watch, below the previous floor of four records (27 September, also a Sunday). A one-bar bar chart carries no distributional information and is retained only so the issue's figure set is complete and comparable in the rollups.

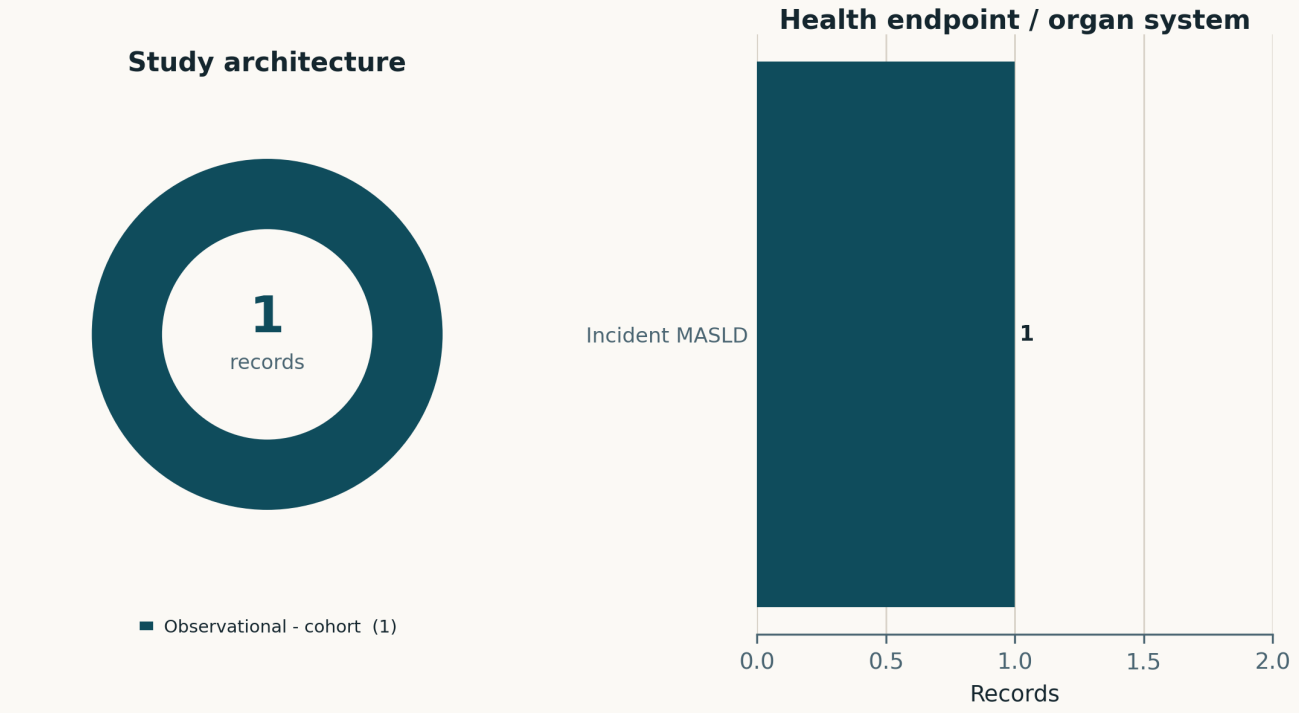



**Fig.** Left — architecture: one prospective cohort. Right — one health endpoint, incident metabolic dysfunction-associated steatotic liver disease. No metadata-only record this issue, the first zero since 4 October.

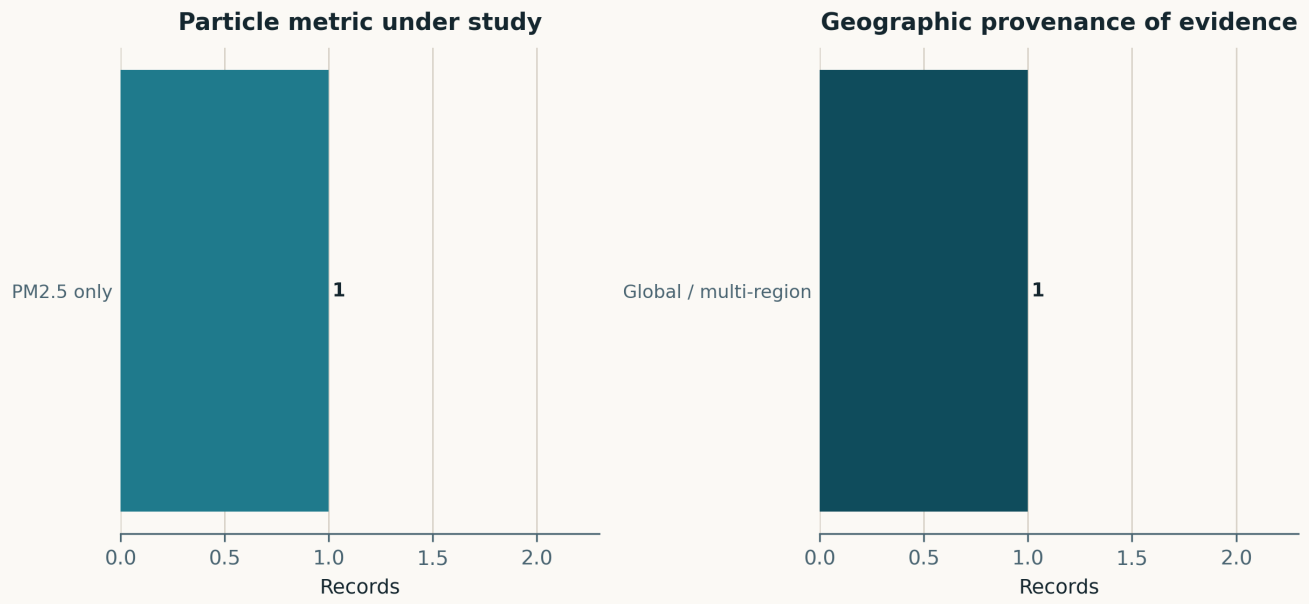



**Fig.** Left — particle metric: $PM_{2.5}$ only, as an annual mean from an interpolated surface. Right — geography: the cohort's country is not stated in the abstract, so the record is assigned to global / multi-region rather than guessed at from cohort size or follow-up.

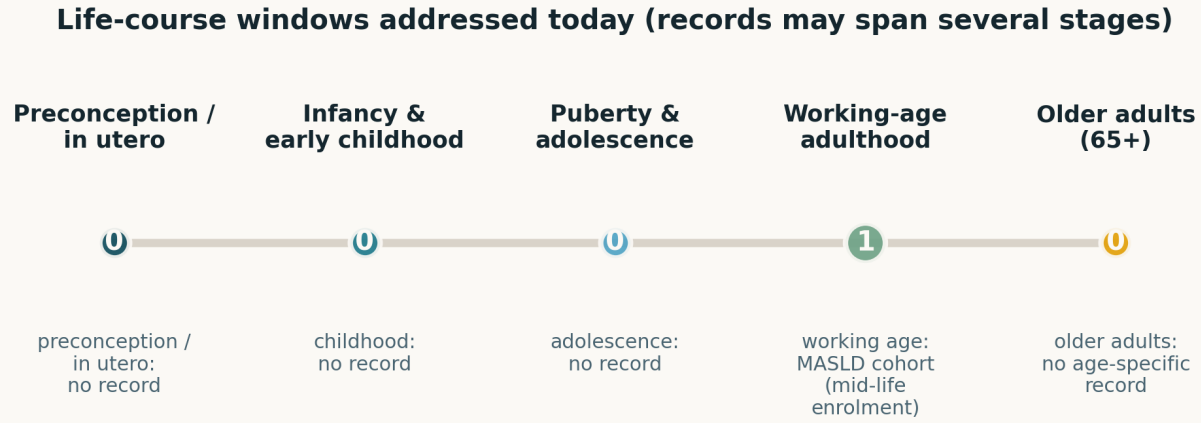



**Fig.** Life-course windows: one record, working-age adulthood (mid-life enrolment, 11.8-year follow-up to incident liver disease). Every other window is empty.

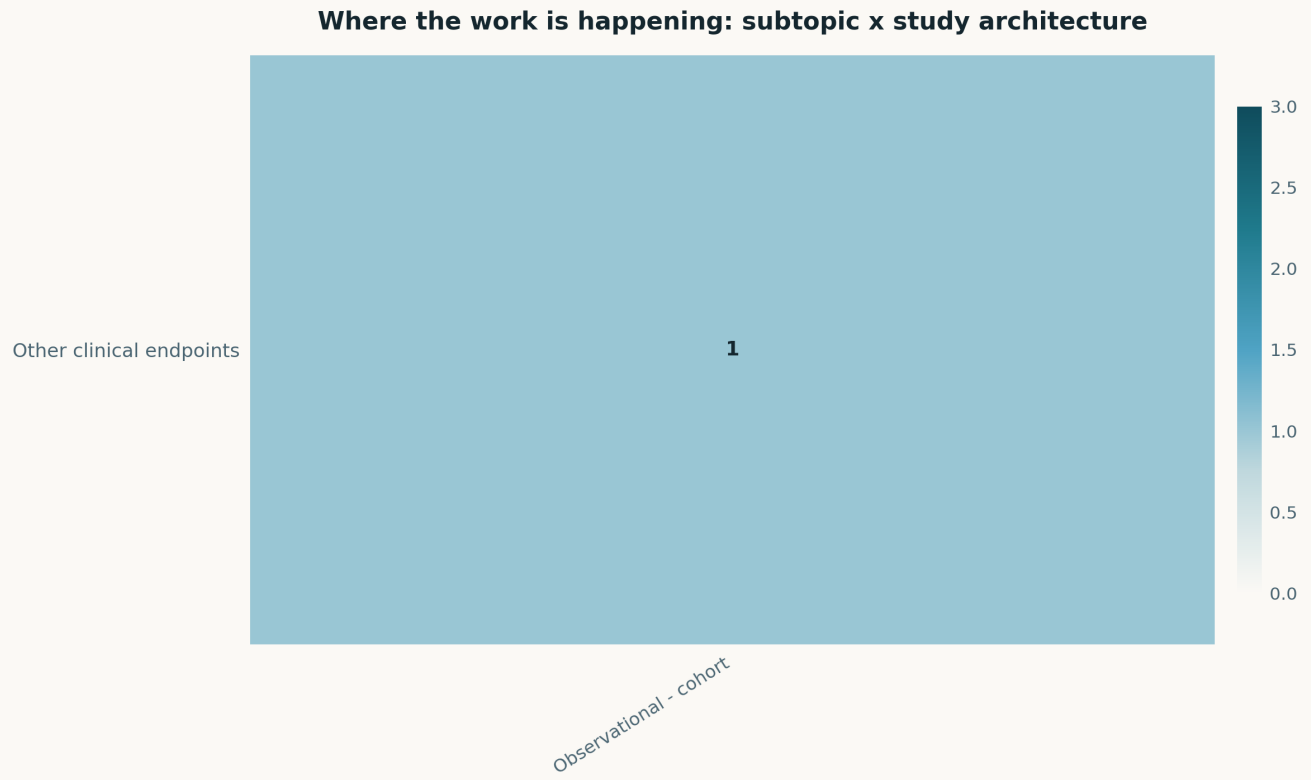



**Fig.** Subtopic × study architecture: a single occupied cell. Shown for continuity with the rest of the series and for the pooled rollups, where this issue contributes one cell of the week.

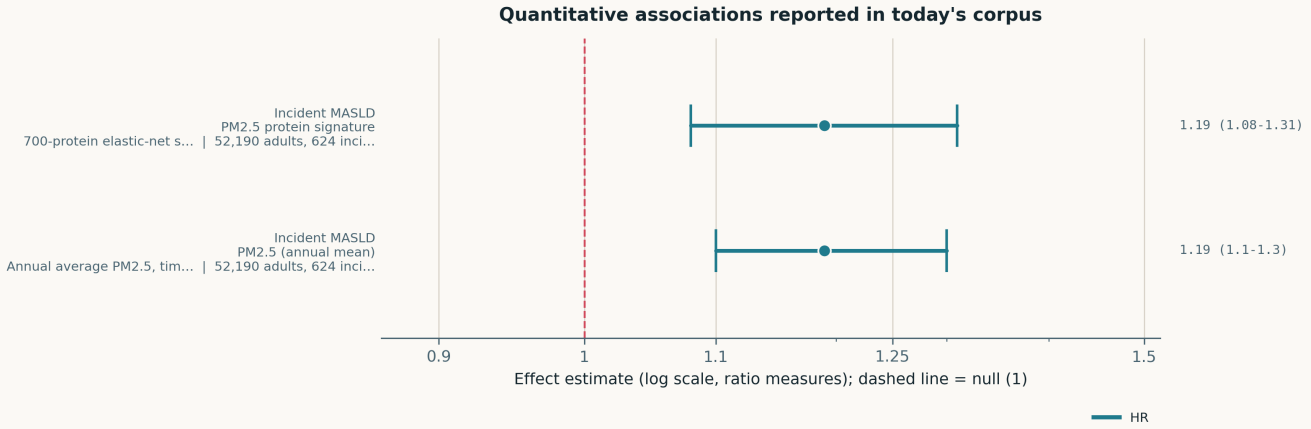



**Fig.** Two hazard ratios with 95 % CIs from one cohort, one exposure model and one set of 624 events. They are not independent estimates: the protein signature is fitted to $PM_{2.5}$ in these same participants, so its near-identical hazard ratio reflects how well the signature reproduces the exposure variable, not a second line of evidence for the association. Reading the pair as mutual confirmation would be a mistake; the panel is shown because the issue's only quantitative content is here.

## Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

### Other clinical endpoints — 1 record

**Ran $PM_{2.5}$ MASLD proteins 2026** J Adv Res

*Association of protein signature of fine particulate matter with MASLD: the underlying pathways and Mendelian randomization*

52,190 participants free of metabolic dysfunction-associated steatotic liver disease at baseline, 624 incident cases over a median 11.8 years. $PM_{2.5}$ gave HR 1.19 (1.10-1.30) and a 700-protein elastic-net signature of $PM_{2.5}$ gave HR 1.19 (1.08-1.31), the proteins enriched in NF-kB signalling and cytokine-receptor interaction; Mendelian randomisation supported CD28, CSF3 and HAVCR2 as causally associated with MASLD, and the signature mediated 15% (2.7-37.7%) of the $PM_{2.5}$-MASLD association. Limitation: a 700-protein signature trained on the same cohort in which it is then tested will absorb variance from everything $PM_{2.5}$ correlates with - smoking, adiposity, deprivation - so the mediated fraction is an upper bound, and its CI nearly spans an order of magnitude; the MR arm addresses protein-to-MASLD causality, not exposure-to-protein. Exposure is an annual mean from an interpolated surface, which is the weakest term in the chain. Relevance: moderate - liver endpoints are under-studied, and 15% mediation at best leaves the mechanism mostly open.

doi: 10.1016/j.jare.2026.10.021

## Corpus register

| First author | Focus | Journal | Design | Metric | Region |
|---|---|---|---|---|---|
| **Other clinical endpoints** | | | | | |
| Ran $PM_{2.5}$ MASLD proteins | Association of protein signature of fine particulate matter with MASLD: the underlying pathways and Mendelian randomization | *J Adv Res* | Prospective cohort | Annual average $PM_{2.5}$ by bilinear interpolation, time-weighted; 700-protein elastic-net signature | Not stated (large prospective cohort, 52,190 participants) |

### Provenance and method

Entry window **11 October 2026**, single day, continuous with the 10 October issue. Harvester legs: **PubMed** health **9**, sensing **3**; **Europe PMC 0** — no record created in the window, the first zero EPMC day since 25 September; **Crossref by ISSN 6**; **arXiv** no entry dated 11 October; **OpenAlex** HTTP 429. 18 raw records deduplicated to **15**, 4 already carried, **11** fresh. The **PubMed connector** (`[EDAT]` 9–11 October) returned 3 PMIDs dated 11 October, 1 appended — **12** candidates screened, **1 admitted**, **11 rejected** with logged reasons, two of them duplicates of records already carried in the 10 October issue. **No record carried on metadata alone. Consensus** returned 10 calibration hits, **0 new**. **Scholar Gateway** fails with `ACCESS_DENIED` and has done for the whole backfill; it is the designated route for recovering numbers that abstracts omit, and it needs reconnecting at the user's end. **ClinicalTrials.gov**: the 4–11 October window is reported in the W41 weekly. Abstracts remain the copyright of their respective publishers.